

Sumit Chhajer

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EXPERIENCE

Tata Advanced Systems Limited

Mumbai

Software Engineering Intern

Jun 2026 – Present

- Developing SPIDER, a full stack security and surveillance platform for defense infrastructure, focusing on real time asset monitoring and high throughput data processing.
- Designing scalable backend architectures and APIs in a Linux environment to ensure low latency system performance and optimized concurrent data pipelines.

Webgile Solutions

Nagpur, India

Data Analyst Intern (Python & SQL)

Dec 2025 – Jan 2026

- Engineered and optimized data transformation pipelines using Python (Pandas, NumPy) and SQL, processing large datasets for statistical analysis and system reliability.
- Constructed complex SQL queries to track KPIs, utilizing data driven methods and statistical concepts to support actionable quantitative decision making.

PROJECTS

High Performance Remote Desktop Architecture

- Architected a production grade remote desktop clone utilizing Rust and WebSockets to facilitate low latency, high throughput concurrent data transmission.
- Designed a robust signaling server architecture matching top tier repositories to ensure seamless real time connectivity, bypassing basic web methods for high performance execution.

DeepHider Steganography & Cryptography Engine

- Engineered a custom 32 bit bit plane slicing algorithm in C/C++ to losslessly embed high resolution image payloads with optimal computational efficiency.
- Implemented a secure XOR stream cipher with SHA 256 key expansion, focusing on object oriented programming principles, memory usage optimization, and execution speed.

IoT Based Smart Helmet for Delivery Workforce Safety

- Designed and developed an IoT enabled smart helmet using Raspberry Pi 4 and ESP32 to enhance rider safety through alcohol monitoring, drowsiness detection, and fall detection architectures.
- Integrated Firebase Realtime Database, GPS tracking, and Twilio API for real time safety monitoring, low latency emergency alerts, and smart vehicle concurrent access control.

EDUCATION

Thadomal Shahani Engineering College

Mumbai

B.E. in Electronics and Telecommunication; CGPA: 8.20 / SGPA: 9.14

Aug 2023 – 2027

- **Relevant Coursework:** Data Structures and Algorithms, Object Oriented Programming (C++), Database Management Systems

Nahata College

2021 – 2023

XII Grade HSC

Tapti Public School

2021

X Grade CBSE

SKILLS & INTERESTS

- **Languages:** C/C++, Python, SQL, Rust, VHDL, JavaScript, HTML/CSS
- **Libraries & Frameworks:** Scikit learn, Pandas, NumPy, WebSockets, OpenCV, MediaPipe, Firebase

- **Tools & Infrastructure:** Linux, Git/GitHub, Docker, AWS, Xilinx ISE, TCP/UDP Protocols
- **Core Competencies:** Object Oriented Programming (OOP), Data Structures & Algorithms, Low Latency Trading Infrastructure, High Throughput Architectures, Concurrent Systems
- **Achievements & Leadership:**
 - **LeetCode:** Solved 100+ algorithmic and data structure optimization problems, sharpening complex programming logic.
 - **Technical Leadership:** Technical Head @ ISTE TSEC (2025–2026); spearheaded university hackathons and advanced engineering workshops.
 - **Standardized Testing:** Achieved a 96.14 percentile ranking in the MHT CET, showcasing strong analytical and quantitative proficiency.
 - **Hackathons & Exhibitions:** Secured 2nd Runner Up at Techathon (out of 90+ teams) and 2nd Runner Up at the TSEC Hardware Project Exhibition.